Activity E2 – Part I

Name and surnames:

Answer the following questions. Concision and accuracy in responses are valued.

1. Why it is impossible (or, at least, extremely difficult) to implement a clock synchronization protocol that provides **zero accuracy** (that is, perfect synchronization)?

2. Give an advantage of Berkeley's synchronization protocol over Cristian's.

3. We have verified that in summer, due to the heat, the drift of our system clock increases. In order to keep its deviation bounded, should we increase or decrease the **synchronization period**? Why?

4. Give a possible negative consequence that the adjustment interval is shorter than the calculated deviation (for example, adjusting an advance of 1 minute in an interval of 30 seconds).

5. Given the following values in a synchronization with the NTP symmetric mode between two nodes A and B, what is the **deviation** and **accuracy** obtained in this synchronization? Consider *min*=5*ms*.

Time of sending *m* by node A	Time of reception of *m* by node B	Time of sending *m'* by node B	Time of reception of *m'* by node A
10h00:00,000	10h00:04,010	10h00:04,030	10h00:00,060

6. A system uses a 64-bit register to store time. The system clock updates this register every microsecond. What is the **resolution** of the system clock? Reason the answer.